

# API Documentation for Holistic Traffic Prediction

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## Introduction

The METR-LA Dataset contains information regarding the traffic conditions through 207 sensors across the city of Los Angeles. This API allows users to send data to a deployed Machine Learning model that makes a prediction of traffic levels based on the data provided by the client.

### Base URL

The API is hosted at:

Base URL: <http://127.0.0.1:5001/>

Method: **GET**

Request: No parameters or body

Response: **"Model Loaded Successfully!"**

The API allows users to do one of the following tasks:

- Check Model Health
- Make a Prediction using the Model

### 1) Test Model Load

URL: <http://127.0.0.1:5001/health>

Method: **GET**

Request: No parameters or body

Response: **"Model Healthy!"**

### 2) Predict using Model

URL: <http://127.0.0.1:5001/predict>

Method: **POST**

Content-Type: **application/json**

**Request:**

The input to the POST method, which makes a prediction using our model, is supposed to be an array of size  $[(1, 10, 207)]$ . This is essentially the traffic predictions across 10 consecutive timesteps for each of the 207 sensors.

Specifically, our model expects a JSON input.

An example input would look something like this:

```
{
  "input" : [
    [ 1, 2, 3, 4, ... 207 columns],
    [ 1, 2, 3, 4, ... 207 columns],
    [ 1, 2, 3, 4, ... 207 columns],
    ... 10 rows ...  ]
  ]
}
```

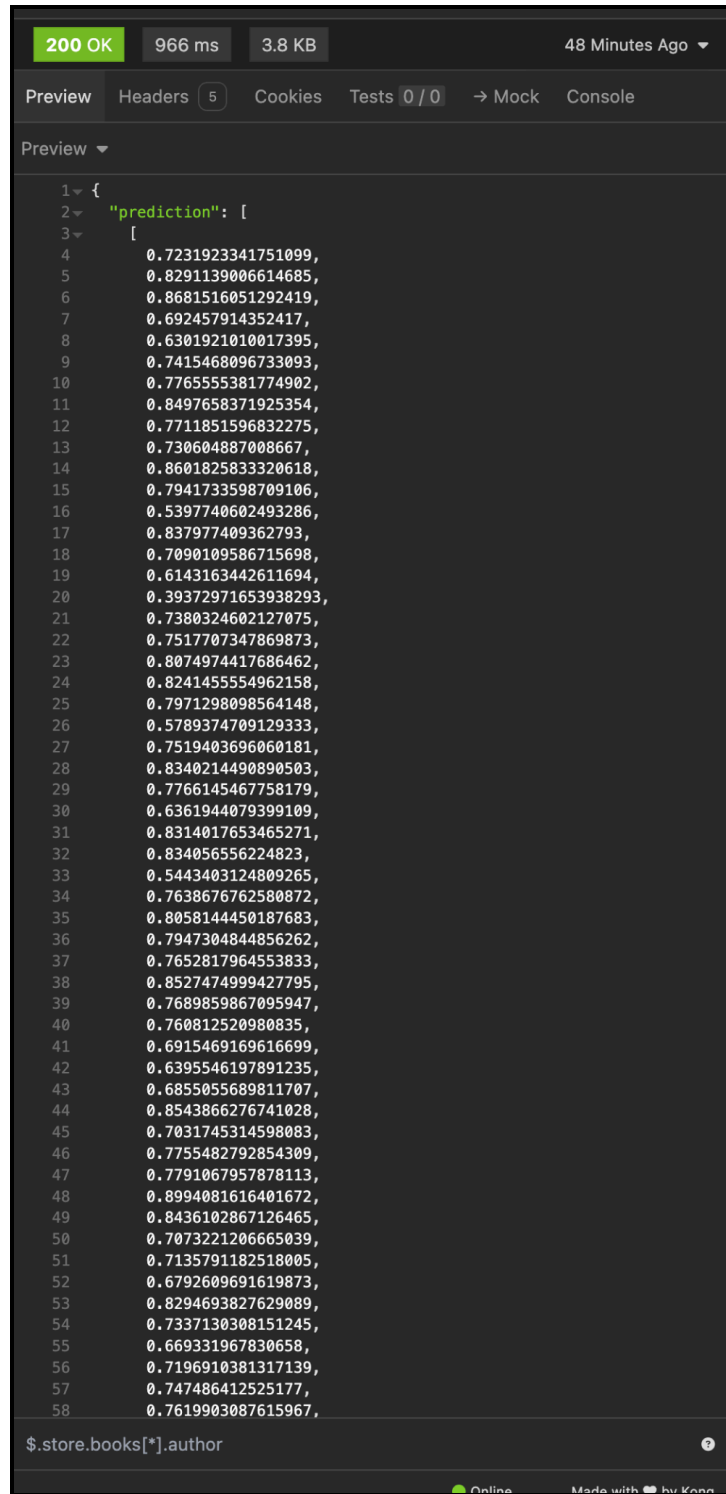
The shape of the above array is  $[(1, 10, 207)]$

**Response:**

The response would be an array of size  $[(1, 1, 207)]$ . This is essentially the prediction of holistic traffic conditions at the next time step.

An example output would look something like this:

```
{
  "prediction" : [
    [ 1, 2, 3, 4, ... 207 columns]
  ]
}
```



## Versioning

Version v1: Updated October 10th, 2024

Thank You!